

$$A := \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix}$$

Spočítejte singulární rozklady matic $A, A^T, AA^T, A^T A$.

Začneme pro A^T :

$$(A^T)^* \cdot A^T = A \cdot A^T = \begin{pmatrix} 7 & -2 & -2 \\ -2 & 7 & -2 \\ -2 & -2 & 7 \end{pmatrix}$$

k této matici nalezneme následující vlastní čísla a vektory:

$$\lambda_a = 3, \quad v_{a1} = \frac{1}{\sqrt{3}} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\lambda_b = 9, \quad v_{b1} = \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad v_{b2} = \frac{1}{\sqrt{6}} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

$$\rightarrow D_{A^T} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix} \quad V_{A^T}^* := \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$A^{T*} = D_{A^T}^* \cdot V_{A^T}^{**} \cdot U_{A^T}^* \rightarrow \begin{pmatrix} -\frac{3}{\sqrt{2}} & \frac{3}{\sqrt{6}} & 1 & 0 \\ 0 & -\frac{6}{\sqrt{6}} & 1 & 0 \\ \frac{3}{\sqrt{2}} & \frac{3}{\sqrt{6}} & 1 & 0 \end{pmatrix} \cdot U_{A^T}^* = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -\frac{3}{\sqrt{2}} & \frac{3}{\sqrt{6}} & 1 & 0 & | & 1 & 1 & 1 & -2 \\ 0 & -\frac{6}{\sqrt{6}} & 1 & 0 & | & 1 & 1 & -2 & 1 \\ \frac{3}{\sqrt{2}} & \frac{3}{\sqrt{6}} & 1 & 0 & | & 1 & -2 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} -\frac{3}{\sqrt{2}} & \frac{3}{\sqrt{6}} & 1 & 0 & | & 1 & 1 & 1 & -2 \\ 0 & -\frac{6}{\sqrt{6}} & 1 & 0 & | & 1 & 1 & -2 & 1 \\ 0 & \frac{6}{\sqrt{6}} & 2 & 0 & | & 2 & -1 & 2 & -1 \end{pmatrix} \sim$$

$$\begin{pmatrix} -\frac{3}{\sqrt{2}} & \frac{3}{\sqrt{6}} & 1 & 0 & | & 1 & 1 & 1 & -2 \\ 0 & -\frac{6}{\sqrt{6}} & 1 & 0 & | & 1 & 1 & -2 & 1 \\ 0 & 0 & 3 & 0 & | & 3 & 0 & 0 & 0 \end{pmatrix} \approx \begin{pmatrix} -\frac{3}{\sqrt{2}} & \frac{3}{\sqrt{6}} & 0 & 0 & | & 0 & 1 & 1 & -2 \\ 0 & -\frac{6}{\sqrt{6}} & 0 & 0 & | & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 0 & | & 1 & 0 & 0 & 0 \end{pmatrix} \approx$$

$$\left(\begin{array}{cccc|cccc} -\frac{3}{\sqrt{2}} & 0 & 0 & 0 & 0 & \frac{3}{2} & 0 & -\frac{3}{2} \\ 0 & -\frac{3}{\sqrt{6}} & 0 & 0 & 0 & \frac{1}{2} & -1 & \frac{1}{2} \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \end{array} \right) \approx \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 & 0 & 0 & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & 0 & 1 & 0 & \frac{1}{\sqrt{3}} & 0 & 0 & 0 \end{array} \right)$$

$$\rightarrow U_{A^T} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \end{pmatrix}$$

(poslední sloupec je vybraný tak, aby U_{A^T} byla unitární)

z $A^T = U_{A^T} \cdot D_{A^T} \cdot V_{A^T}^*$ dostaneme zbylé rozklady:

$$A = V_{A^T} \cdot D_{A^T}^* \cdot U_{A^T}^*$$

$$A \cdot A^T = V_{A^T} \cdot D_{A^T}^* \cdot U_{A^T}^* \cdot U_{A^T} \cdot D_{A^T} \cdot V_{A^T}^* = V_{A^T} \cdot D_{A^T}^* \cdot D_{A^T} \cdot V_{A^T}^*$$

$$A^T \cdot A = U_{A^T} \cdot D_{A^T} \cdot V_{A^T}^* \cdot V_{A^T} \cdot D_{A^T}^* \cdot U_{A^T}^* = U_{A^T} \cdot D_{A^T} \cdot D_{A^T}^* \cdot U_{A^T}^*$$

takže:

$$A = U_A \cdot D_A \cdot V_A^* = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} \cdot \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \end{pmatrix}$$

$$A^T = U_{A^T} \cdot D_{A^T} \cdot V_{A^T}^* = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & -\frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \end{pmatrix} \cdot \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$A \cdot A^T = U_{A \cdot A^T} \cdot D_{A \cdot A^T} \cdot V_{A \cdot A^T}^* = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} \cdot \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 3 \end{pmatrix} \cdot \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$\begin{aligned}
A^T \cdot A &= U_{A^T \cdot A} \cdot D_{A^T \cdot A} \cdot V_{A^T \cdot A}^* = \\
&= \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & -\frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \end{pmatrix} \cdot \begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \end{pmatrix}
\end{aligned}$$